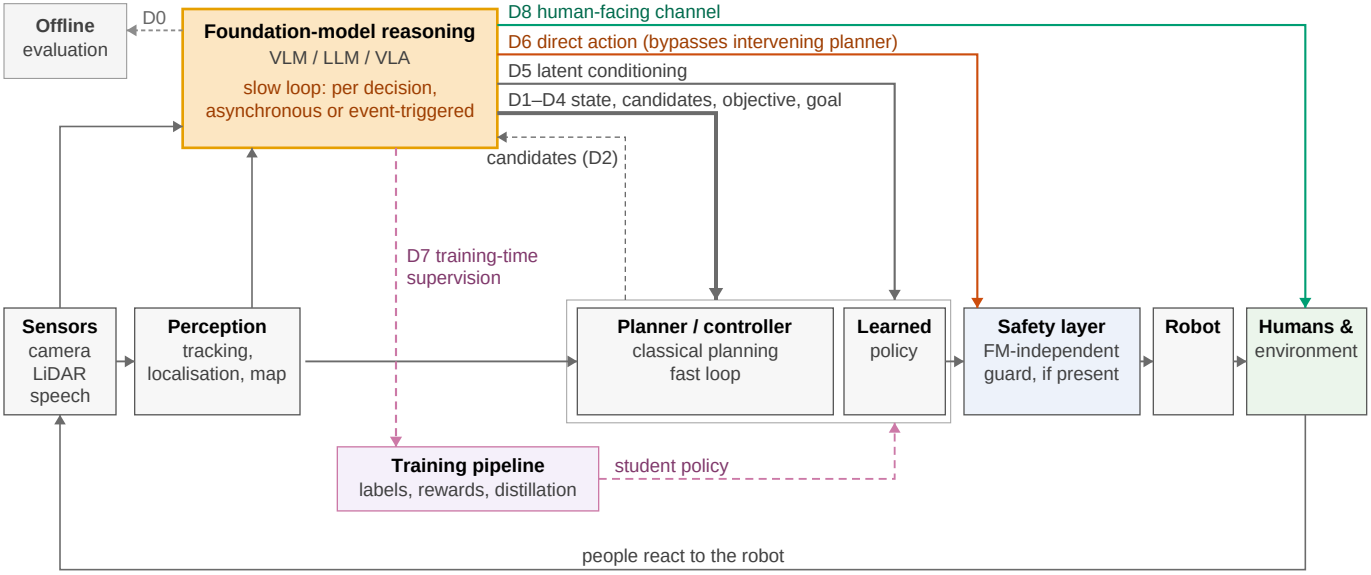
Offline
evaluation
D0
Foundation-model reasoning
VLM / LLM / VLA
slow loop: per decision,
asynchronous or event-triggered
D8 human-facing channel
D6 direct action (bypasses intervening planner)
D5 latent conditioning
D1–D4 state, candidates, objective, goal
candidates (D2)
D7 training-time
supervision
Sensors
camera
LiDAR
speech
Perception
tracking,
localisation, map
Planner / controller
classical planning
fast loop
Learned
policy
Safety layer
FM-independent
guard, if present
Robot
Humans &
environment
Training pipeline
labels, rewards, distillation
student policy
people react to the robot